OPEN SOURCE · AI · LEGAL TECH

LegalLens

AI-powered contract risk analyzer.
Understand what you sign — in minutes, for free.

Multi-Agent AI | RAG + Hybrid Search | 100% Free Stack | PDF / DOCX

THE PROBLEM

Contracts are risky by default

Most people sign without truly understanding the legal exposure hidden in plain text.

Dense legal language

Opaque terminology, nested conditions, and buried exceptions that non-lawyers miss.

Legal review is expensive

Attorney review costs $300–$500/hr — unaffordable for freelancers, startups, and individuals.

Time pressure

Counterparties push for quick signatures — rarely enough time to read every clause carefully.

Hidden risk clauses

Non-compete, IP assignment, and auto-renewal clauses carry serious consequences when overlooked.

THE SOLUTION

A multi-agent AI pipeline

Upload a contract. Get clause-by-clause risk analysis with plain-language explanations and suggested rewrites.

Extract Clauses → Classify Risk → Retrieve Precedents → Generate Recommendations

ARCHITECTURE

End-to-end pipeline

Upload
PDF / DOCX

›

Parse
PyMuPDF / docx

›

Extract
13 clause types

›

Classify
Low / Med / High

›

Retrieve
Hybrid RAG

›

Recommend
Plain language

›

Report
PDF / HTML

LangGraph orchestration

Celery + Redis async queue

Real-time progress

CORE FEATURES

Everything you need. Nothing you don't.

Clause Extraction

Detects 13 legal clause types — liability, IP assignment, non-compete, auto-renewal — with confidence scores.

Risk Classification

Chain-of-thought assigns Low / Medium / High risk with a 0–1 score and full explanation per clause.

Precedent Retrieval

Hybrid RAG (dense + BM25) searches a legal knowledge base to ground every recommendation.

Plain-Language Recs

Human-readable explanations with suggested alternative clause language — ready to negotiate.

TECH STACK

Modern. Open source. Free.

AI / LLM

LangChain + LangGraph
Ollama · Groq · Gemini
sentence-transformers

VECTOR SEARCH

ChromaDB (dense)
BM25 (sparse)
Hybrid re-ranking

BACKEND

Python 3.12 · FastAPI
Celery + Redis
Pydantic v2

DOCUMENT PARSING

PyMuPDF (PDF)
python-docx (DOCX)
Chunked extraction

FRONTEND

Next.js 14 · TypeScript
Tailwind CSS · shadcn/ui

DEVOPS

Docker Compose
GitHub Actions CI
pytest · ruff · mypy

LLM PROVIDER STRATEGY

One env var. Three free options.

OLLAMA — LOCAL

- ✓ No internet needed
- ✓ Unlimited requests
- ✓ Complete privacy
- ✗ Slower (local GPU/CPU)

GROQ — FREE TIER

- ✓ Very fast inference
- ✓ 30 req/min free
- ✓ Llama 3.1 70B
- ✗ Rate limits apply

GEMINI — FREE TIER

- ✓ Gemini 2.0 Flash
- ✓ Strong reasoning
- ✓ Free daily quota
- ✗ Daily quota cap

Embeddings always local via sentence-transformers

ENGINEERING CHALLENGES

Real problems. Concrete fixes.

CHALLENGE

Free-tier rate limits

Groq's 6,000 TPM cap caused 429 errors when classifying clauses in parallel.

✓ FIX

Configurable `LLM_REQUEST_DELAY` + sequential processing. Zero delay for local Ollama.

CHALLENGE

Inconsistent LLM output

LLMs returned reasoning as arrays, invented clause type names, or produced malformed JSON.

✓ FIX

Pydantic v2 validators coerce lists → strings and unknown enums → `OTHER`. Max 2 retries.

CHALLENGE

Orchestrator data flow bug

Classifier and retriever ran in parallel — but retriever needed classified clauses first.

✓ FIX

Refactored LangGraph pipeline: classify first, then retrieve precedents using classified output.

RESULTS & IMPACT

From upload to insight. In minutes.

13	3	$0	3
clause types detected	risk tiers with reasoning	API cost (fully free)	LLM providers supported

End-to-end working pipeline | PDF + HTML reports | Async Celery queue | Next.js 14 UI

OPEN SOURCE

Built in public. Open to all.

LegalLens is open source — contributions, forks, and feedback are welcome.

github.com/ahmadghalawinji/legallens

View on GitHub

Contact

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